THE STABILIZER SITS 2-CARTESIANLY OVER THE INERTIA STACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original definition 044Q: sources/stacks-chow/tags.txt, source lines 5409–5409. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 0448: sources/stacks-chow/tags.txt, source lines 5391–5391. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 04ZN: sources/stacks-chow/tags.txt, source lines 6471–6471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06PB: sources/stacks-chow/tags.txt, source lines 8565–8565. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Assume $B \to S$ and (U, R, s, t, c) as in Definition 044Q (1). Let G/U be the stabilizer group algebraic space of the groupoid (U, R, s, t, c, e, i), see Definition 0448. There is a canonical 2-cartesian diagram*

$$\begin{array}{ccc} \mathcal{S}_G & \longrightarrow & \mathcal{S}_U \\ \downarrow & & \downarrow \\ \mathcal{I}_{[U/R]} & \longrightarrow & [U/R] \end{array}$$

of stacks in groupoids of $(Sch/S)_{fppf}$.

Proof. By Lemma 04ZN it suffices to prove that the morphism $s' : R' \to G$ of Lemma 06PB isomorphic to the base change of s by the structure morphism $G \to U$. This base change property is clear from the construction of s'. □

Date: stacks-project:spaces-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.